

Lab 3 – Architecture Selection and Justification

NAME-ARPITA PILLAI

SRN-PES2UG24AM028

CLASS-5A(AIML)

Architecture Selection:

We chose Layered Architecture for the Self-Service Coffee Kiosk System.

Architectural Choice and Justification:

Layered Architecture is appropriate for the kiosk because it separates the system into Presentation, Business, and Data layers. The Presentation Layer contains the User Interface Component for touch-screen interaction. The Business Layer contains the Order Manager, Payment Service, and Receipt Printer components for order processing, payment handling, and receipt printing. The Data Layer contains the Menu Database Component for storing menu and pricing information.

Reason 1 – Separation of Responsibilities:

The architecture provides a clear separation between user interaction, business processing, and data storage. This makes the system easier to understand and maintain because each component has a focused responsibility. For example, the User Interface handles customer selections while the Order Manager coordinates order processing and communication with the payment, receipt, and database components.

Reason 2 – Suitability for the Kiosk:

The coffee kiosk has a focused set of functions: selecting a coffee type and size, processing credit-card payment, printing receipts, and accessing menu and pricing information. A layered structure is therefore suitable because it provides a simple and organized design without the additional operational complexity associated with independently deployed microservices.

Security Advantage:

Layered separation can help protect system data by preventing the user interface from directly accessing the database or payment-processing functionality. Requests can pass through the appropriate business components, allowing access to payment and stored menu information to be controlled at the business layer.

Performance Benefit:

The layered design provides a straightforward processing path between the user interface, business components, and menu database. For this focused kiosk application, this avoids unnecessary distributed-service communication and supports efficient processing of customer orders.